

UI Binding Transfer for Bone-driven Facial Rigs

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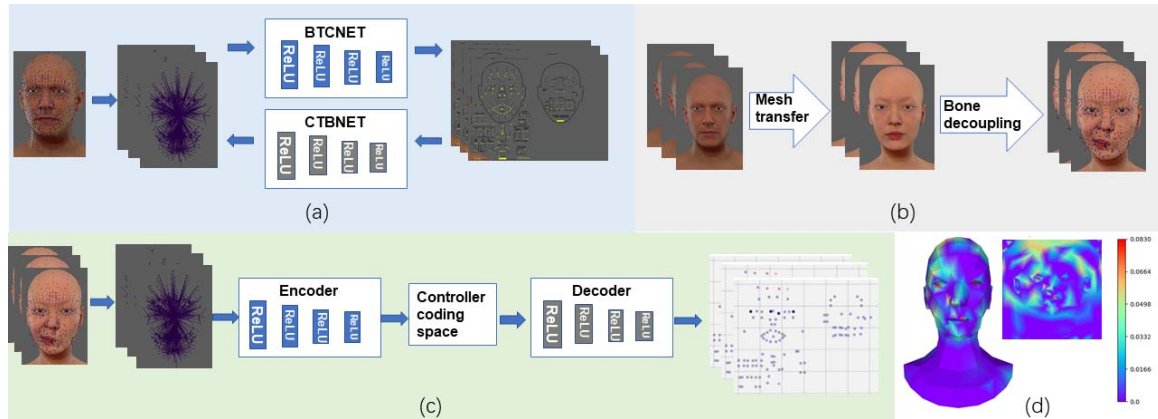


Figure 1: UI binding transfer system. (a) UI binding and inverse-binding function learning; (b) Target data acquisition; (c) Target binding function training; (d) Visible error plot on 3d space and UV space.

ABSTRACT

We propose an automatic method to transfer the UI binding from a rigged model to a new target mesh. We use feed forward neural networks to find the mapping functions between bones and controllers of source model. The learned mapping networks then become the initial weights of an auto-encoder. Then the auto-encoder is retrained using target controllers-bones pairs obtained by the mesh transfer and bones decoupling method. Our system only requires neutral expression of the target person but allows artists to customize other basic expressions, and is evaluated by the semantic reproducibility of basic expressions and the semantic similarity.

Index Terms: Computing methodologies—Computer graphics—Graphics systems and interfaces—Virtual reality; Computing methodologies—Computer graphics—Animation—Procedural animation;

1 INTRODUCTION

Mesh rigging is the foundation of virtual human production and driving a virtual human to generate expressions is a necessary part of facial animation. Bone-driven method is now used by animators as the first choice for facial modeling thanks to its independence to animate flexible expressions. Bone-driven facial rigs are often augmented with 2D UI controllers, which allows intuitive control for artists to pose faces. However, creating those controllers and binding them to the rig requires much manual effort. Thus given a rigged model such as Metahuman [1], we propose an automatic UI binding transfer method and we define the mapping from controllers to bones as UI binding and the mapping from bones to controllers as UI inverse-binding.

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Bailey et al [2] first proposed binding solution for human body. They divided skin deformation into linear and nonlinear parts and the nonlinear part was fitted by neural network. But their binding skipped the bones, which was not suitable for the bone-driven facial model. Holden et al. [3] proposed a gaussian process regression(GPR) combined with Jacobian inverse optimization method to establish the mapping function from bones to controllers and introduced feed forward neural networks(FF Net) later [4]. But their methods omitted the rotation angle of bones and just fit one specific model, thus they can't handle the 'transfer' problem. Kim et al. [5] created their own UI layouts for different models but artists needed to relearn how to pose expressions.

Our main contributions are the following three aspects. Firstly, we learn UI binding and inverse-binding from the source rigged model automatically by using feed forward neural network. Secondly given a target neutral mesh, target bones data is automatically obtained by accelerated mesh transfer and bones decoupling method and the size of target bones training data is only 1/1000 of the size of source bones training data. Finally we simultaneously obtain the UI binding and inverse-binding of the target model by training an auto-encoder. Our system also allows artists to additionally specify bones data of target model's basic expressions while the semantics can be reproduced after the binding transfer.

2 METHOD

UI binding and inverse-binding function learning UI binding is fitted by BTCNET, where inputting bones data and outputting controllers data. UI inverse-binding is fitted by CTBNET, where inputting controllers data and outputting bones data. BTCNET and CTBNET both have 4-5 fully connected layers and take Relu as activation function. Bone's local rotation angles are converted to world coordinate of three unit vectors by rotation matrices. Thus together with bone's world coordinate, each bone is represented by a 12 dimension tensor independently of each other. 2D controllers are represented by their local coordinate to keep local correlation. Thus each controller is represented by a 2 dimension tensor. Total

dimension of bones data is $12b_n$ and total dimension of controllers data is $2c_n$, where b_n represents bone's number and c_n represents controller's number. Our experiment shows that, the number of neurons of the i th layer N_i obeys the equation(1) performing best on fitting performance, where l_n is the number of layers.

$$N_i = \frac{i}{l_n - 1} (c_n + b_n) \quad (1)$$

Target data acquisition Given a rigged model, we automatically sample the UI panel by uniform distribution within the motion range to get an animation of source mesh. Then given a neutral mesh of target model, we apply accelerated mesh transfer [6] for each frame to get an animation of target mesh. Finally combining bones position of source neutral mesh, we decouple bones position for each frame by accelerated bone decoupling [7] method. In this way, we automatically get target controllers-bones pairs.

Target binding function training Target binding and inverse-binding function is trained by an auto-encoder(AE). The encoder is initialized by the BTCNET and thus has the same structure with BTCNET while the decoder is initialized by the CTBNET. Inputting target bones data $\mathbf{B}_t$ into the encoder, we get verified controllers data $\mathbf{C}_t'$. And inputting verified controllers data into the decoder, we get verified bones data $\mathbf{B}_t'$. Finally, a constraint is added as equation(2) shows, where $\mathbf{C}_t$ represents target controllers data. The ratio of the training set to the validation set is 8 to 2.

$$RMSEloss = \sqrt{\frac{1}{b_n} \sum_{i=1}^n \|\mathbf{B}_t - \mathbf{B}_t'\|_2^2} + \sqrt{\frac{1}{c_n} \sum_{i=1}^n \|\mathbf{C}_t - \mathbf{C}_t'\|_2^2} \quad (2)$$

3 RESULTS AND DISCUSSION

We compare our BTCNET with Holden's GPR [3] and FF Net [4] as shown in Figure 2. Although we get a mean squared loss of 0.007 cm^2 nearly close to 0.006 cm^2 of Holden's, their methods obviously ignore the importance of angles and get semantically wrong result.

Besides, we compare our CTBNET with artists by computing the root mean squared loss(RMSE) between source bones data and verified bones data. Verified bones data is obtained by inputting the same controllers, when posing source model to get source bones data, into CTBNET. An error plot can be seen in figure 1(d) and the RMSE is 0.0623cm for 15 bones and 0.0943cm for 215 bones while the world distance at the x direction of two corners of the model's mouth is 4.2068cm, which means our work is effective to replace manual labor.

After the transfer, we wish to get 'semantic similar' expressions between source and target model, as shown in figure 3. It's too subjective to evaluate what extent can be defined as 'similar', thus we change the thought. Since the Metahuman [1] series admittedly have 'semantic similar' expression when same controllers are set, we judge the 'semantic similar' by computing the RMSE between the source and verified Metahuman controllers data. Verified controllers data is obtained by inputting target Metahuman bones data corresponding to the source controllers data into the decoder of the AE. The RMSE is 0.1835 cm for 15 bones and 0.2052 cm for 215 bones when the local coordinate of controllers is about 2.5cm on average, which shows that our method can meet the need of semantic similarity.

4 CONCLUSION

We present an automatic binding transfer system that can achieve fine control of virtual faces with accurate semantic reproducibility and similarity. Users only need to offer a neutral mesh of the target virtual human and the target bones training data is generated automatically. What's more, we also allow artists to add manually defined basic meshes as reference of the 'true' expressions of the target person and we can reproduce these expressions precisely. Our



Figure 2: A comparison of UI binding effect. (a) Original expression; (b) Holden's GPR [3]; (c) Holden's feed-forward network [4]; (d) Our method.



Figure 3: The semantically similar expression after transfer.

system will greatly improve the efficiency during digital human expression production, promote the use of digital human in various industries, and provide the underlying key technology support for mass production of digital assets.

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